GNG/ELG 5301 — Professional Skills and Responsibility
Team Research Project Guidelines

Professor: Mohammad Alnabhan **University of Ottawa**
Course Theme: Professional Skills and Responsibilities in AI-Enabled Organizations

1. Project Overview

The central activity of GNG/ELG 5301 is an **integrated multidisciplinary team research project**.

Rather than completing a collection of unrelated assignments, each team will identify an approved professional problem related to technology and AI-enabled organizations, investigate it using credible scholarly and professional evidence, develop a literature-informed analysis, and produce practical recommendations for responsible professional practice.

The project is designed to help you practise the skills that professionals and project teams use in real organizations:

- defining a meaningful problem;
- identifying stakeholders and their interests;
- establishing scope and objectives;
- finding and evaluating credible evidence;
- conducting a structured literature review;
- synthesizing information rather than simply summarizing sources;
- managing a multidisciplinary team;
- planning and monitoring project work;
- identifying risks and limitations;
- communicating with technical and non-technical audiences;
- making evidence-based professional recommendations;
- documenting decisions and contributions;
- using generative AI responsibly;
- presenting findings professionally.

The project is intentionally **research-oriented but not a traditional academic research thesis**. Your objective is to produce a defensible, evidence-based professional analysis that could be useful to an organization, professional community, technology practitioner, or decision-maker.

2. Umbrella Theme

Professional Skills and Responsibilities in AI-Enabled Organizations

All projects must fit within the broad theme of professional skills and responsibilities in organizations increasingly affected by artificial intelligence and other technology-enabled systems. Possible areas include:

- responsible AI governance;
- privacy and accountability;
- cybersecurity responsibilities;
- intellectual property and AI;
- fairness and workplace decision-making;
- AI-assisted software development;
- AI and project management;
- professional responsibility in technology-enabled work;
- professional skills required for AI-enabled organizations;
- responsible use of generative AI;
- AI-related risks in business and professional environments;
- human oversight and accountability;
- AI literacy and professional decision-making;
- technology, trust, and organizational responsibility.

These are examples rather than a list of mandatory topics. Teams may propose another topic if it clearly fits the course theme and can support a rigorous literature-informed investigation.

3. What the Project Is, and Is Not

This is an applied research project. Your team will:

1. define a professional problem;
2. formulate a focused research/project question;
3. establish a clear scope;
4. search for relevant scholarly and professional evidence;
5. evaluate the quality and relevance of sources;
6. organize evidence systematically;
7. identify themes, patterns, disagreements, risks, and gaps;
8. develop a defensible analysis;
9. connect the analysis to professional responsibilities;
10. formulate practical recommendations;
11. communicate the results through a professional report and presentation.

This is NOT primarily a programming project. Programming is optional. A team may develop a small notebook, visualization, simple computational analysis, prototype, or other technical artifact when it strengthens the research question. However, Advanced programming is not required and will not be rewarded simply because it is technically complex. The quality of the project will be judged primarily on the quality of the question, evidence, analysis, professional reasoning, transparency, recommendations, communication, and understanding of limitations.

4. Research Orientation

Teams will use a common research process involving:

- a defined research question;
- a documented search strategy;
- a search log;
- source-screening decisions;
- a structured evidence table;
- common or clearly defined evidence/coding categories;
- transparent synthesis;
- explicit limitations;
- evidence-based recommendations.

This common structure makes projects more comparable and makes the resulting evidence easier to evaluate.

5. Research Ethics and Human Participants

This course project is **not a human-participant research study**. For the purposes of this course, teams must **not** collect:

- surveys;
- interviews;
- focus-group data;
- observational data;
- personal information;
- identifiable participant data;

for course credit unless the instructor separately authorizes an ethics-compliant process. Do not recruit participants or collect personal data simply because you believe it would improve the project. Your project should normally be completed using existing scholarly literature, professional reports, publicly available information, approved datasets, documented cases, technical documentation, standards, and other appropriate secondary evidence.

6. Team Structure

Students will work in **stable multidisciplinary teams of five**. Teams are assigned after a skills profile to encourage diversity in:

- disciplinary background;
- technical experience;
- business experience;
- research experience;
- communication skills;
- professional interests.

You are not expected to have identical skills. The purpose of the multidisciplinary structure is to make the team operate more like a professional project team in which members contribute different strengths.

7. Team Charter

Early in the semester, each team will create a one-page team charter. The charter is a professional working agreement that gives the team a clear process for dealing with predictable problems. The charter should establish:

- team members and roles;
- preferred communication channel;
- meeting frequency;
- expected response time;
- method for making decisions;
- method for resolving disagreements;
- expectations for contribution;
- procedure for addressing missed deadlines;
- escalation procedure when a problem cannot be resolved internally.

Teams must maintain an evidence-of-contribution record throughout the project. A team mark is therefore **not automatically an equal individual mark**. Where there is evidence of substantially unequal contribution, the instructor may adjust an individual student's mark.

8. Project Workflow

Stage 1: Team Formation and Topic Exploration

Teams will: review their collective skills; identify areas of interest; examine possible project directions; discuss professional relevance; identify potential stakeholders; select a preliminary topic. **Enrol your group on Brightspace by the deadline**.

Stage 2: Problem Definition and Scope

Teams will convert a broad topic into a specific professional problem. A strong problem definition should answer:

- What problem are we investigating?
- Why does it matter?
- Who is affected?
- In what context does it occur?
- What is inside our scope?
- What is outside our scope?
- What professional decision or practice does this problem affect?

Avoid questions that are so broad that they cannot be investigated meaningfully within one semester. **Weak example:** What are the effects of AI? **Stronger example:** What professional accountability mechanisms should organizations establish when generative AI is used in software-development workflows? The second question establishes a clearer professional context and creates a basis for evidence, analysis, and recommendations.

9. Three-Minute Technical Pitch: 2%, In-class

Each team will deliver a concise progress pitch. The pitch should communicate:

1. the professional problem;
2. why the problem matters;
3. the proposed research/project question;
4. the intended scope;
5. the anticipated contribution.

The purpose is a test of whether the team can explain its proposed project clearly and professionally. Pitch format: **3 minutes,** followed by instructor/audience questions.

10. Research Skills and Evidence **In lecture and In tutorials**

The project requires students to learn how to locate and evaluate credible evidence. Teams should use an appropriate combination of sources, potentially including:

- peer-reviewed journal articles **and** conference papers;
- academic books or chapters;
- government publications, and authoritative institutional publications;
- professional standards and recognized professional organizations;
- credible industry reports;
- technical documentation.

The quality of a source matters more than simply increasing the number of references. Teams should be able to explain: Why is this source credible, relevant, and appropriate for the claim we are making?

11. Evidence Map / Progress Clinic **In tutorial**

Each team will bring:

- a search log;
- inclusion/exclusion decisions;
- an evidence table containing approximately 5–12 credible sources;
- emerging themes;
- one project question for TA feedback.

The evidence map is a working research tool. It is not simply a bibliography. A useful evidence table might include **Source, Type, Research focus, Key finding, Relevance, Limitation, Theme.** Teams should begin identifying relationships among sources. For example:

- Do several sources agree?
- Where do sources disagree?
- Are findings context-dependent?
- What risks repeatedly appear?
- What professional responsibilities are emphasized?
- What evidence is missing?
- What assumptions do different sources make?

12. Team Research and Project Proposal: 16%

The proposal is the team's formal plan for completing the project. It is a 5–7 pages, excluding references and appendices where appropriate, and should have:

1. Problem Definition

- the professional problem;
- the context;
- why it matters;
- affected stakeholders.

2. Research / Project Question

State one focused primary question. Supporting subquestions may be used where necessary.

3. Scope

Clearly identify:

- what the team will investigate;
- what it will not investigate;
- relevant assumptions;
- boundaries of the project.

4. Preliminary Evidence

Present the most important early evidence and explain how it supports the proposed investigation.

5. Research / Search Approach

Explain:

- where the team will search;
- search terms or search strategy;
- source-selection criteria;
- inclusion/exclusion principles;
- how evidence will be organized and synthesized.

6. Expected Contribution

Explain what the completed project is expected to provide. The contribution may be:

- a professional framework;
- evidence synthesis;
- decision guidance;
- recommendations;
- risk-management guidance;
- comparison of approaches;
- responsible-practice recommendations;
- a small technical or analytical artifact.

7. Team Roles

Identify responsibilities and how roles will be coordinated.

8. Timeline

Show major tasks and milestones through final submission.

9. Risk Register

Identify important project risks and mitigation strategies. Example:

Risk	Likelihood	Impact	Mitigation
Topic becomes too broad	Medium	High	Narrow scope around defined context
Insufficient scholarly evidence	Medium	High	Expand databases/search terms
Unequal contribution	Medium	High	Weekly contribution tracking
Technical analysis becomes too complex	Medium	Medium	Prioritize research question over coding complexity

10. References: Use an appropriate professional/academic citation format consistently.

13. Data Literacy and Responsible-AI Brief: 7%

All teams will complete guided data-literacy and responsible-AI activities using a **common small, non-sensitive dataset**.

14. From Evidence to Recommendations

A major purpose of the project is to move beyond literature summary.

A weak report says: Source A says X. Source B says Y. Source C says Z. A stronger report asks: What does the combined evidence tell us? An excellent report goes further: Given the evidence, professional responsibilities, risks, limitations, and organizational context, what should a practitioner or organization do?

15. Final Research Report and Professional Briefing: 30%

The final submission is a **12–15-page professional research report**, excluding references and appendices. The report should be written for an informed professional audience rather than only for the course instructor.

Required structure

1. Executive Summary: Briefly communicate (like a paper abstract):

- the problem;
- the research question;
- major findings;
- key implications;
- most important recommendations.

A decision-maker should be able to understand the project's main message from the executive summary.

2. Introduction and Problem Context: Explain:

- the problem;
- its context;
- significance;
- stakeholders;
- project objectives.

3. Research Question and Scope: Clearly state the question and boundaries of the investigation.

4. Method / Evidence Process: which documents:

- search strategy;
- source-selection process;
- evidence organization;
- analytical approach;
- any coding/categories used.

5. Literature / Evidence Synthesis: Do not organize this section as a sequence of source summaries. Instead, organize the discussion around:

- themes;
- patterns;
- relationships;
- disagreements;
- risks;
- evidence gaps.

6. Analysis: Interpret the evidence in relation to the project question. This is the most important intellectual component of the report. The analysis should demonstrate judgment rather than simply repeat what sources say.

7. Professional Responsibility Implications: Discuss relevant responsibilities such as:

- ethics;
- privacy;
- security;
- accountability;
- intellectual property;
- equity/fairness;
- professional obligations;
- organizational/public impact.

Not every project will require equal treatment of every responsibility. Focus on the responsibilities that are genuinely relevant to your problem.

8. Recommendations: Provide actionable recommendations. Where appropriate, distinguish between immediate actions; medium-term actions; longer-term considerations.

9. Limitations: Identify limitations honestly. Examples include evidence gaps, limited scope, source limitations, assumptions, dataset limitations, methodological constraints, generalizability concerns. A limitation does not weaken a project when it is appropriately recognized and explained.

10. References: Use consistent scholarly/professional citation (APA style).

11. Evidence Appendix: Include supporting evidence tables or other documentation where appropriate.

16. Final Team Presentation and Facilitation: 10%

A Presentation is a maximum of 10 minutes, followed by Q&A. **Its grade isn't negotiable**.

17. Responsible Use of Generative AI

Generative AI may be used only within the boundaries specified for each assessment. Where permitted, AI may support activities such as:

- brainstorming;
- language editing;
- format checking;
- generating alternative ways of explaining an idea.

However, AI **cannot** replace:

- student analysis;
- source evaluation;
- professional judgment;
- interpretation;
- authorship.

Students remain responsible for every submitted claim. You must verify:

- factual claims;
- citations;
- quotations;
- calculations;
- references;
- data interpretations.

AI-generated references must never be assumed to be real or accurate.

AI-use statement

When AI use is permitted and materially affects submitted work, include a brief statement identifying tool used; date; purpose; how the output was verified or revised. Undeclared or inappropriate AI use may be treated as an academic integrity concern.

18. What Counts as a Strong Project?

A strong project demonstrates five things simultaneously:

1. A focused problem: The team knows exactly what it is investigating.
2. Credible evidence: The evidence is appropriate, relevant, and accurately documented.
3. Genuine synthesis: The team identifies relationships among sources rather than producing a collection of summaries.

4. Professional judgment: The team considers responsibility, risks, stakeholders, limitations, and competing considerations.
5. Actionable communication: The final product tells the audience what the evidence means and what should be done.

19. What Does Not Make a Project Strong?

The following do **not** automatically produce a high-quality project:

- using a complicated machine-learning model;
- writing a large amount of code;
- collecting a large number of sources;
- producing a very long report;
- adding unnecessary technical terminology;
- creating many charts without interpretation;
- using generative AI to produce large amounts of text;
- presenting information source-by-source without synthesis.

The project is evaluated on **quality of thinking and professional execution**, not technical complexity for its own sake.

20. Final Checklist

Before submitting the final project, confirm that your team can answer **yes** to the following:

Problem

- Is our problem clearly defined?
- Is the scope realistic?
- Have we identified relevant stakeholders?

Research

- Is our research question focused?
- Is our search process documented?
- Are our sources credible?
- Have we evaluated source quality?
- Have we synthesized evidence rather than summarized sources?

Analysis

- Does our analysis answer the research question?
- Are claims supported by evidence?
- Have we considered alternative interpretations?
- Have we identified limitations?

Professional Responsibility

- Have we considered the responsibilities relevant to our problem?
- Have we addressed risks and potential impacts?
- Are our recommendations professionally defensible?

Recommendations

- Are our recommendations specific?
- Are they realistic?
- Are they supported by our findings?
- Is it clear who should act and what they should do?

Communication

- Can a non-specialist understand our main message?
- Is the report professionally organized?
- Are visuals readable and cited?
- Is the executive summary useful?

Teamwork

- Can we demonstrate individual contributions?
- Did we respond to feedback?
- Did we manage roles and deadlines professionally?

AI and Academic Integrity

- Have we verified every citation and factual claim?
- Have we followed the course AI-use requirements?
- Have we included an AI-use statement where required?
- Have we avoided uploading confidential or restricted information to public AI tools?

21. The Standard to Aim For

The project should ultimately answer one question:

> **Can your team investigate a real professional problem, make sense of credible evidence, exercise responsible professional judgment, and communicate a defensible course of action to others?**

That is the central purpose of the GNG/ELG 5301 team project.

Prof. Mohammad Alnabhan

** End of Document **